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Real-Time Driver Drowsiness Detection System
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Abstract
Driver fatigue remains a leading contributing factor to vehicular accidents worldwide. Often, the cognitive decline associated with drowsiness occurs before the driver is consciously aware of their impairment, necessitating an automated, non-invasive monitoring solution. This report details the development and implementation of a real-time, camera-based drowsiness detection system designed to evaluate facial cues and intervene when prolonged eye closure indicates potential sleepiness. The system processes live video feeds at the frame level, utilizing MediaPipe for high-fidelity facial landmark detection. By calculating the Eye Aspect Ratio EAR for both eyes, the software tracks the duration of eye closures against a configurable temporal threshold. Should a driver's eyes remain closed beyond the established baseline, the system initiates immediate visual warnings and an audible Pygame-driven alarm, incorporating a cooldown mechanism to mitigate alarm fatigue. The architecture operates entirely on local hardware using Python, OpenCV, and NumPy, thereby eliminating the need for continuous cloud connectivity or intrusive wearable sensors. Furthermore, a Flask-based web interface serves as the primary presentation layer for real-time monitoring. Software verification confirms that the core state-machine logic functions as specified, accurately differentiating between normal physiological blinking and dangerous microsleeps based on a calibrated EAR threshold of 0.21.

Keywords: Driver Drowsiness, Eye Aspect Ratio, MediaPipe, OpenCV, Real-Time Detection, Flask.

1. Introduction
Automotive safety is fundamentally dependent on the continuous alertness and reaction capability of the driver. However, drowsiness progressively degrades this alertness, often exhibiting subtle early physiological indicators that drivers themselves may fail to recognize. A

fatigued driver typically demonstrates an increased blink rate, prolonged eyelid closures, and a notable reduction in micro-corrections to steering, all while subjectively perceiving themselves as capable of operating the vehicle. By the time the impairment manifests in erratic vehicle behavior, the margin for corrective action is frequently insufficient to prevent a collision.

A primary challenge in mitigating fatigue-related accidents is the difficulty of continuously measuring alertness in a non-invasive manner. Conventional Advanced Driver Assistance Systems (ADAS), such as lane departure warnings, are fundamentally reactive; they detect the consequences of driver inattention rather than the cognitive state of the driver. These systems intervene only after a critical vehicular error has been committed.

Computer vision provides a proactive methodology to address this deficiency. By leveraging standard optical sensors, it is possible to continuously monitor the driver's facial expressions and eye states without requiring intrusive biometric wearables. The ocular state, visible in every video frame, serves as a highly reliable proxy for overall cognitive alertness.

This research focuses on engineering a computationally efficient, automated system that processes webcam video locally. The proposed architecture is designed to identify potential drowsiness by analyzing prolonged eye closures and to alert the driver via simultaneous visual and auditory stimuli. Rather than deploying computationally expensive deep learning classification models for every frame, this system employs a geometric, time-series approach, ensuring high performance on standard hardware.

The primary objectives of this project are defined as follows:

- To capture and process real-time vehicular video streams efficiently using the OpenCV library.
- To accurately detect facial landmarks and isolate the ocular regions irrespective of minor head movements.
- To implement a mathematical model, specifically the Eye Aspect Ratio (EAR), to quantify the degree of eye openness.
- To engineer a robust, time-based state machine capable of distinguishing normal blinking patterns from hazardous eye closures.
- To generate timely, state-dependent visual warnings and auditory alarms.
- To construct a responsive web-based monitoring dashboard utilizing the Flask framework.

The efficacy of such a system relies on its determinism and adaptability across varying lighting conditions and user profiles, principles which guided the architectural decisions detailed in the subsequent methodology.

2. Methodology

2.1 System Architecture and Workflow

The system operates as a continuous, frame-by-frame processing pipeline. The workflow initiates by capturing live optical data via a standard webcam, which is passed to OpenCV for initial frame manipulation and color space conversion. MediaPipe is subsequently utilized to perform high-speed inference, mapping complex facial landmarks. Upon isolating the coordinates of the eyes, the Eye Aspect Ratio is calculated. If the EAR falls below a designated threshold for a sustained duration, a state transition is triggered, identifying the subject as drowsy. Finally, the system initiates a Pygame audio alert and updates the Flask-served frontend.

2.2 Technology Stack

To ensure the system remains lightweight and executes in real time without specialized GPU hardware, the technology stack was carefully optimized:

● Python: Core algorithmic logic and process execution
● OpenCV Optical capture and primary frame processing
● MediaPipe: High-speed, real-time facial landmark detection
● NumPy: Matrix calculations for EAR metric extraction
● Flask: Backend routing for the monitoring dashboard
● HTML/JS/CSS Frontend presentation and user interface controls
● Pygame: Asynchronous triggering of the auditory alarm

2.3 The Eye Aspect Ratio EAR and State Logic

The Eye Aspect Ratio is a scalar value that estimates eye openness independent of the subject's distance from the camera. Utilizing six specific landmarks per eye, the EAR is calculated. As the eyelids close, the vertical distances approach zero, causing a sharp decline in the ratio. The system averages the EAR of both eyes and applies an exponential moving average (smoothing factor of 0.65 to filter out single-frame tracking noise.

To prevent false positives from natural blinking (typically lasting 0.1 to 0.4 seconds), the system utilizes a strict state machine. In the NORMAL state, EAR values remain above the 0.21 threshold. If the EAR remains below threshold for 0.9 seconds or more, the system transitions to WARNING. If the closure persists for 1.8 seconds or more, it shifts to DROWSY, triggering the alarms. Should the face leave the frame for 0.5 seconds or more, a FACE NOT DETECTED state pauses the timer.

3. Results and Discussion

3.1 System Verification and Functional Testing

The primary validation phase focused on verifying the mathematical integrity of the

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